

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{324} \frac{1}{m_\gamma^2} c_\gamma^2 (\overline{y_u}^{p13} (y_u^{p14} (-7g_1^2 + 6g_3^2) \delta_{i1i2} - 18g_3^2 y_u^{p12} \delta_{i1i4}) + \right. \\
& \left. \overline{y_u}^{p11} (-18g_3^2 y_u^{p14} \delta_{i2i3} + y_u^{p12} (-81s_\gamma^2 \overline{y_u}^{r13} y_u^{r14} + (-7g_1^2 + 6g_3^2) \delta_{i3i4})) \right) + \\
& \frac{1}{18} g_3^4 (3\delta_{i1i4} \delta_{i2i3} - \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[m_3] + \frac{1}{12} g_3^4 (3\delta_{i1i4} \delta_{i2i3} - \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_3] + \\
& \frac{4}{45} g_3^4 (-3\delta_{i1i4} \delta_{i2i3} + \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_3] + \\
& \frac{1}{486} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (16g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[m_d^p] - \\
& \frac{5}{1296} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (16g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_d^p] + \\
& \frac{2}{1215} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (16g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_d^p] + \\
& \frac{8}{81} \sum_p g_1^4 \text{LF}_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{27} \sum_p g_1^4 \text{LF}_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{32}{405} \sum_p g_1^4 \text{LF}_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{4}{81} \sum_p g_1^4 \text{LF}_{3,0}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \\
& \frac{5}{54} \sum_p g_1^4 \text{LF}_{4,-1}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \frac{16}{405} \sum_p g_1^4 \text{LF}_{5,-2}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{243} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (4g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[m_q^p] - \\
& \frac{5}{648} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (4g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_q^p] + \\
& \frac{4}{1215} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (4g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_q^p] + \\
& \frac{1}{486} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (64g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[m_u^p] - \\
& \frac{5}{1296} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (64g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_u^p] + \\
& \frac{2}{1215} \sum_p (27g_3^4 \delta_{i1i4} \delta_{i2i3} + (64g_1^4 - 9g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_u^p] + \\
& \frac{1}{54} c_\gamma^2 (\overline{y_u}^{p13} (y_u^{p14} (-2g_1^2 + 3g_3^2) \delta_{i1i2} - 9g_3^2 y_u^{p12} \delta_{i1i4}) + \\
& \overline{y_u}^{p11} (-9g_3^2 y_u^{p14} \delta_{i2i3} + y_u^{p12} (27s_\gamma^2 \overline{y_u}^{r13} y_u^{r14} + (-2g_1^2 + 3g_3^2) \delta_{i3i4}))) \text{LF}_{1,2}[m_\Phi] + \\
& \frac{1}{36} c_\gamma^2 (-2g_1^2 \overline{y_u}^{p13} y_u^{p14} \delta_{i1i2} + \overline{y_u}^{p11} y_u^{p12} (9c_\gamma^2 \overline{y_u}^{r13} y_u^{r14} - 2g_1^2 \delta_{i3i4})) \text{LF}_{2,1}[m_\Phi] + \\
& \frac{1}{162} g_1^2 (9c_\gamma^2 \overline{y_u}^{p13} y_u^{p14} \delta_{i1i2} + (9c_\gamma^2 \overline{y_u}^{p11} y_u^{p12} + 8g_1^2 \delta_{i1i2}) \delta_{i3i4}) \text{LF}_{3,0}[m_\Phi] - \\
& \frac{5}{54} g_1^4 \text{LF}_{4,-1}[m_\Phi] \delta_{i1i2} \delta_{i3i4} + \frac{16}{405} g_1^4 \text{LF}_{5,-2}[m_\Phi] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{81}{41} g_1^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{27} g_1^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{32}{405} g_1^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_u^{i1}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_u^{i1}] + \\
& \frac{1}{243} g_1^2 (-9g_3^2 \delta_{i1i4} \delta_{i2i3} + (-8g_1^2 + 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_u^{i1}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_u^{i1}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_u^{i2}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_u^{i2}] + \\
& \frac{1}{243} g_1^2 (-9g_3^2 \delta_{i1i4} \delta_{i2i3} + (-8g_1^2 + 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_u^{i2}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_u^{i2}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_u^{i3}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_u^{i3}] + \\
& \frac{1}{243} g_1^2 (-9g_3^2 \delta_{i1i4} \delta_{i2i3} + (-8g_1^2 + 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_u^{i3}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_u^{i3}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_u^{i4}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_u^{i4}] + \\
& \frac{1}{243} g_1^2 (-9g_3^2 \delta_{i1i4} \delta_{i2i3} + (-8g_1^2 + 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_u^{i4}] + \\
& \frac{1}{486} g_1^2 (9g_3^2 \delta_{i1i4} \delta_{i2i3} + (8g_1^2 - 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_u^{i4}] + \\
& \frac{1}{1296} (-9g_3^4 \delta_{i1i4} \delta_{i2i3} + g_3^2 (64g_1^2 + 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_3, m_u^{i1}] + \\
& \frac{1}{1296} (-9g_3^4 \delta_{i1i4} \delta_{i2i3} + g_3^2 (64g_1^2 + 3g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_3, m_u^{i1}] + \\
& \frac{1}{1296} (-225g_3^4 \delta_{i1i4} \delta_{i2i3} + g_3^2 (-128g_1^2 + 75g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_3, m_u^{i1}] + \\
& \frac{1}{162} (9g_3^4 \delta_{i1i4} \delta_{i2i3} + g_3^2 (8g_1^2 - 3g_3^2)$$